American International University-Bangladesh (AIUB)
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Disaster Relief Management System

A Software Engineering Project
Submitted By Group-03

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1. PROJECT PROPOSAL

1.1 Background to the Problem

i. Disaster Situation in Bangladesh

Bangladesh is globally recognized as a nation highly susceptible to environmental hazards. Its position within the Ganges-Brahmaputra-Meghna delta, characterized by a low-lying landscape and its border with the Bay of Bengal, significantly increases its risk of natural calamities. Records from the Bangladesh Disaster Management Bureau indicate that the country has faced over 200 major disaster events in the last thirty years. The scale of disasters is massive, with millions being displaced, injured, livelihoods destroyed, and lives lost in disasters such as Cyclone Sidr in 2007, Cyclone Aila in 2009, and floods in Sylhet in 2022, where millions were displaced within days of the disaster occurring. The cost of disaster management in Bangladesh is very high.

ii. Existing Challenges and Project Context

Even though Bangladesh has improved in disaster preparedness, relief operations still depend a lot on manual systems. This leads to several problems:

- **Lack of Centralized Information Management:** The people in need are registered using paper forms at each of the camps, and there is no common database. This causes issues such as double registration and loss of information. The decision makers at the higher levels do not have access to timely and accurate information about the number of people being sheltered and the availability of resources.
- **Challenges For Impacted Populations:** Individuals affected by disasters frequently face significant obstacles, including slow response times in aid distribution and critical shortages of fundamental necessities such as safe drinking water, healthcare, and emergency housing. Further, there is confusion about registration, inequality of distribution of relief, and limited access to information regarding available aid.
- **Resource Misallocation and Duplication:** Without a coordinated system, different groups can end up sending the same items to the same places, such as food and medicine, while critical items such as clean water and emergency medications may not be sent to hard-to-reach areas.
- **Absence of Donor Transparency:** Donors often can't clearly see how their money and aid are used.

To solve these challenges, the Disaster Relief Management System (DRMS) is proposed as a centralized online web app that facilitates unified coordination among administrators, volunteers, victims, and contributors in one accessible space. It will offer immediate information and assist in coordinating relief efforts more efficiently.

iii. Target Group of Users

1. Administrators:

Administrators such as government officials and district coordinators oversee all relief operations. They register camps, allocate resources, and manage volunteers.

2. Volunteers:

Individuals who engage in voluntary service, including medical practitioners, engineers, and community support personnel, undertake missions in the field.

3. Affected Persons:

Those affected are the main beneficiaries. The displaced families and those injured who need food, shelter, and medical care are the main beneficiaries. They sign up to receive aid, get their own digital ID, and can track their aid.

4. Donors:

Donors such as individuals, companies, people living abroad, and international organizations who support relief efforts. Using the DRMS platform, they can easily see verified needs, donate safely online, and track how their money is being used.